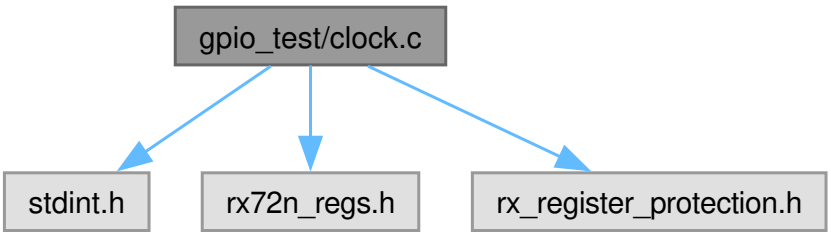
>>

>>

>>

```
#include <stdint.h>
#include "rx72n_regs.h"
#include "rx_register_protection.h"
```

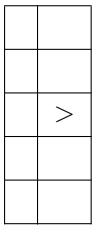


*

>>

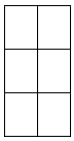
>>>

enum clock_byte_constants_t : uint8_t



```
00075             : uint8_t {
00076     k_hococr_running   = 0x00U, /**< HOCOCR: 0 = HOCO running */
00077     k_pllcr2_enable    = 0x00U, /**< PLLCR2: 0 = PLL running (inverted logic) */
00078     k_memwait_one      = 0x01U, /**< MEMWAIT: 1 wait state for ICLK > 120 MHz */
00079     k_oscovfsr_hcovf   = 0x08U, /**< OSCOVFSR bit 3: HOCO stable */
00080     k_oscovfsr_plovf   = 0x04U, /**< OSCOVFSR bit 2: PLL stable */
00081 } clock_byte_constants_t;
```

enum clock_constants_word_t : uint16_t



```
00043             : uint16_t {
00044     /*
00045      * PLLCR layout (16 bits):
00046      *   bits [1:0]  PLIDIV  : 00=/1, 01=/2, 10=/3
00047      *   bit  [4]    PLLSRCSEL: 0=MOSC, 1=HOCO
00048      *   bits [13:8] STC     : multiply = (STC + 1) / 2
00049      *
00050      * HOCO 16 MHz / 1 * 12 = 192 MHz PLL output.
00051      * STC = (12 * 2) - 1 = 23 = 0x17.
00052      * PLLSRCSEL = 1 (HOCO) -> bit 4 set.
00053      */
00054     k_pllcr_hoco_x12 = 0x1710U,
00055
00056     /*
00057      * SCKCR2: bits [7:4] UCK = /(N+1), bits [3:0] reserved (must be 0x1).
00058      * UCK = 0x3 -> /4 -> 192 MHz / 4 = 48 MHz exact.
00059      */
00060     k_sckcr2_uck_div4 = 0x0031U,
00061
00062     /* PACKCR: bit 0 UPLLSEL -- 0 routes UCLK divider input from main PLL. */
00063     k_packcr_pll_source = 0x0000U,
00064 } clock_constants_word_t;
```

enum clock_extra_addrs_t : uintptr_t

```
00036             : uintptr_t {
00037     k_packcr_addr = 0x00080044U, /**< Peripheral clock source: 0=PLL, 1=PPLL */
00038 } clock_extra_addrs_t;
```

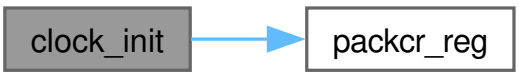
enum clock_sckcr_t : uint32_t

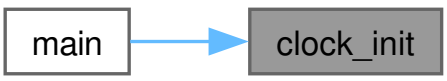
```
00066             : uint32_t {
00067     /*
00068      * SCKCR dividers:
00069      *   FCK=/4 (48), ICK=/2 (96), PSTOP1=1, PSTOP0=1, BCK=/4 (48),
00070      *   PCKA=/2 (96), PCKB=/4 (48), PCKC=/2 (96), PCKD=/2 (96).
00071      */
00072     k_sckcr_value = 0x21C21211U,
00073 } clock_sckcr_t;
```

void clock_init (
                    void)

>>

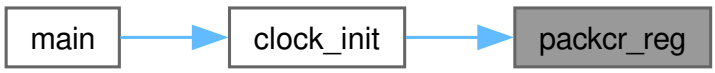
```
00101 {
00102     volatile rx_system_regs_t *sys = system_regs();
00103
00104     /* Step 0: unlock PRC0 (clock) + PRC1 (module stop) */
00105     *prcr_reg() = k_rx_prcr_unlock_prc0_prc1;
00106
00107     /* Step 1: start HOCO if not already running. */
00108     sys->hococr = k_hococr_running;
00109
00110     /* Step 2: wait for HOCO to stabilise. */
00111     while ((sys->oscovfsr & k_oscovfsr_hcovf) == 0U) {
00112         ___asm___ volatile("nop");
00113     }
00114
00115     /* Step 3: program PLL: HOCO 16 MHz / 1 * 12 = 192 MHz, and enable. */
00116     sys->pllcr  = k_pllcr_hoco_x12;
00117     sys->pllcr2 = k_pllcr2_enable;
00118
00119     /* Step 4: wait for PLL to stabilise. */
00120     while ((sys->oscovfsr & k_oscovfsr_plovf) == 0U) {
00121         ___asm___ volatile("nop");
00122     }
00123
00124     /* Step 5: MANDATORY -- raise flash wait state before ICLK exceeds 120 MHz. */
00125     *memwait_reg() = k_memwait_one;
00126
00127     /* Step 6: program all bus dividers. */
00128     sys->sckcr = k_sckcr_value;
00129
00130     /* Step 7: select main PLL as the UCLK divider source (UPLLSEL=0). */
00131     *packcr_reg() = k_packcr_pll_source;
00132
00133     /* Step 8: program the USB clock divider. */
00134     sys->sckcr2 = k_sckcr2_uck_div4;
00135
00136     /* Step 9: switch the system clock source to PLL. */
00137     sys->sckcr3 = k_sckcr3_cksel_pll;
00138
00139     /* Step 10: relock PRCR. */
00140     *prcr_reg() = k_rx_prcr_lock;
00141 }
```





volatile uint16_t * packcr_reg (
void) [inline], [static]

```
00087 {
00088     return (volatile uint16_t *)k_packcr_addr;
00089 }
```

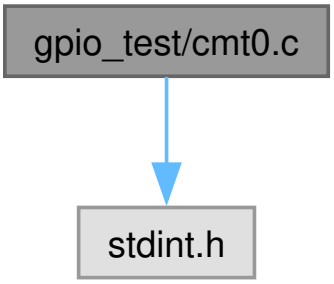



```
00001 /**
00002  * @file clock.c
00003  * @brief RX72N clock initialisation: HOCO 16 MHz -> PLL 192 -> ICLK 96 / PCKB 48.
00004  *
00005  * @details
00006  * This test sources the PLL from the internal HOCO (16 MHz) instead of
00007  * the STAR PCB's 24 MHz external crystal on MOSC/EXTAL. The choice is
00008  * deliberate (isolate the PLL bring-up from crystal startup), not because
00009  * the board lacks a crystal -- it has one.
00010  *
00011  * Target clocks:
00012  *   - PLL   = HOCO 16 MHz x 12 = 192 MHz
00013  *   - ICLK  = 192 / 2  =  96 MHz  (CPU)
00014  *   - PCKA  = 192 / 2  =  96 MHz
00015  *   - PCKB  = 192 / 4  =  48 MHz  (CMT0 ThreadX tick)
00016  *   - PCKC  = 192 / 2  =  96 MHz
00017  *   - PCKD  = 192 / 2  =  96 MHz
00018  *   - FCLK  = 192 / 4  =  48 MHz  (flash)
00019  *   - BCLK  = 192 / 4  =  48 MHz
00020  *   - UCLK  = 192 / 4  =  48 MHz  (not used, but set for completeness)
00021  *
00022  * Path: HOCO 16 MHz -> PLIDIV /1 -> STC x12 -> 192 MHz PLL output.
00023  *
00024  * SPDX-License-Identifier: MIT
00025  * @copyright Copyright (c) 2026 Locked Inc.
00026  */
00027
00028 #include <stdint.h>
00029
00030 #include "rx72n_regs.h"
00031 #include "rx_register_protection.h"
00032
00033 /*
      ============================================================================
00034  * Register addresses not exposed by the rx_system_regs_t struct
00035  *
      ============================================================================
      */
00036 typedef enum : uintptr_t {
00037     k_packcr_addr = 0x00080044U, /**< Peripheral clock source: 0=PLL, 1=PPLL */
00038 } clock_extra_addrs_t;
00039
00040 /*
      ============================================================================
00041  * Clock configuration constants
00042  *
      ============================================================================
      */
00043 typedef enum : uint16_t {
00044     /*
00045      * PLLCR layout (16 bits):
00046      *   bits [1:0]   PLIDIV   : 00=/1, 01=/2, 10=/3
00047      *   bit  [4]     PLLSRCSEL: 0=MOSC, 1=HOCO
00048      *   bits [13:8] STC      : multiply = (STC + 1) / 2
00049      *
00050      * HOCO 16 MHz / 1 * 12 = 192 MHz PLL output.
00051      * STC = (12 * 2) - 1 = 23 = 0x17.
00052      * PLLSRCSEL = 1 (HOCO) -> bit 4 set.
```

```
00053     */
00054     k_pllcr_hoco_x12 = 0x1710U,
00055
00056     /*
00057      * SCKCR2: bits [7:4] UCK = /(N+1), bits [3:0] reserved (must be 0x1).
00058      * UCK = 0x3 -> /4 -> 192 MHz / 4 = 48 MHz exact.
00059      */
00060     k_sckcr2_uck_div4 = 0x0031U,
00061
00062     /* PACKCR: bit 0 UPLLSEL -- 0 routes UCLK divider input from main PLL. */
00063     k_packcr_pll_source = 0x0000U,
00064 } clock_constants_word_t;
00065
00066 typedef enum : uint32_t {
00067     /*
00068      * SCKCR dividers:
00069      *   FCK=/4 (48), ICK=/2 (96), PSTOP1=1, PSTOP0=1, BCK=/4 (48),
00070      *   PCKA=/2 (96), PCKB=/4 (48), PCKC=/2 (96), PCKD=/2 (96).
00071      */
00072     k_sckcr_value = 0x21C21211U,
00073 } clock_sckcr_t;
00074
00075 typedef enum : uint8_t {
00076     k_hococr_running   = 0x00U, /**< HOCOCR: 0 = HOCO running */
00077     k_pllcr2_enable    = 0x00U, /**< PLLCR2: 0 = PLL running (inverted logic) */
00078     k_memwait_one      = 0x01U, /**< MEMWAIT: 1 wait state for ICLK > 120 MHz */
00079     k_oscovfsr_hcovf   = 0x08U, /**< OSCOVFSR bit 3: HOCO stable */
00080     k_oscovfsr_plovf   = 0x04U, /**< OSCOVFSR bit 2: PLL stable */
00081 } clock_byte_constants_t;
00082
00083 /*
    ============================================================================
00084  * Inline accessors
00085  *
    ============================================================================
    */
00086 static inline volatile uint16_t *packcr_reg(void)
00087 {
00088     return (volatile uint16_t *)k_packcr_addr;
00089 }
00090
00091 /*
    ============================================================================
00092  * clock_init -- bring the chip from LOCO 240 kHz to PLL 192/96/48 MHz.
00093  *
00094  * Uses HOCO (16 MHz internal oscillator) as the PLL source.
00095  * No external crystal required.
00096  *
00097  * Must run before any peripheral that depends on PCKB is touched.
00098  * Called from main() before cmt0_init() and before tx_kernel_enter().
00099  *
    ============================================================================
    */
00100 void clock_init(void)
00101 {
00102     volatile rx_system_regs_t *sys = system_regs();
00103
00104     /* Step 0: unlock PRC0 (clock) + PRC1 (module stop) */
00105     *prcr_reg() = k_rx_prcr_unlock_prc0_prc1;
00106
00107     /* Step 1: start HOCO if not already running. */
00108     sys->hococr = k_hococr_running;
00109
00110     /* Step 2: wait for HOCO to stabilise. */
00111     while ((sys->oscovfsr & k_oscovfsr_hcovf) == 0U) {
00112         ___asm___ volatile("nop");
00113     }
00114
00115     /* Step 3: program PLL: HOCO 16 MHz / 1 * 12 = 192 MHz, and enable. */
00116     sys->pllcr  = k_pllcr_hoco_x12;
00117     sys->pllcr2 = k_pllcr2_enable;
00118
00119     /* Step 4: wait for PLL to stabilise. */
00120     while ((sys->oscovfsr & k_oscovfsr_plovf) == 0U) {
00121         ___asm___ volatile("nop");
00122     }
00123
00124     /* Step 5: MANDATORY -- raise flash wait state before ICLK exceeds 120 MHz. */
00125     *memwait_reg() = k_memwait_one;
00126
00127     /* Step 6: program all bus dividers. */
00128     sys->sckcr = k_sckcr_value;
00129
00130     /* Step 7: select main PLL as the UCLK divider source (UPLLSEL=0). */
00131     *packcr_reg() = k_packcr_pll_source;
00132
00133     /* Step 8: program the USB clock divider. */
```

```
00134     sys->sckcr2 = k_sckcr2_uck_div4;
00135
00136     /* Step 9: switch the system clock source to PLL. */
00137     sys->sckcr3 = k_sckcr3_cksel_pll;
00138
00139     /* Step 10: relock PRCR. */
00140     *prcr_reg() = k_rx_prcr_lock;
00141 }
```

#include <stdint.h>



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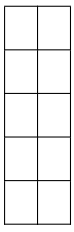
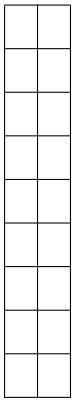
*

enum cmt0_cfg_t : uint16_t



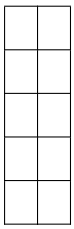
```
00045             : uint16_t {
00046     k_cmt0_cmcor_val       = 14999U,  /**< (48 MHz / 32) / 100 Hz - 1 = 14999 for 100 Hz tick */
00047     k_cmt0_cmcr_val        = 0x0041U, /**< CMCR: CMIE=1 (interrupt), CKS=01 (PCKB/32) */
00048     k_cmstr0_ch0_bit       = 0x0001U, /**< CMSTR0 bit 0: CMT0 start/stop */
00049     k_prcr_unlock_prc0_prc1 = 0xA503U, /**< Write PRCR=0xA503 to enable PRC0+PRC1 writes */
00050     k_prcr_lock            = 0xA500U, /**< Write PRCR=0xA500 to relock PRC0+PRC1 */
00051 } cmt0_cfg_t;
```

enum cmt_addr_t : uintptr_t



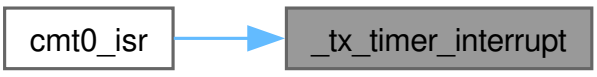
```
00027             : uintptr_t {
00028     k_cmstr0_addr  = 0x00088000U, /**< CMSTR0: shared CMT0/CMT1 start register (Ch 15) */
00029     k_cmt0_cmcr    = 0x00088002U, /**< CMT0 CMCR: control register (Ch 15) */
00030     k_cmt0_cmcnt   = 0x00088004U, /**< CMT0 CMCNT: counter value (Ch 15) */
00031     k_cmt0_cmcor   = 0x00088006U, /**< CMT0 CMCOR: compare-match target (Ch 15) */
00032
00033     k_icu_ir_base  = 0x00087000U, /**< ICU.IR[] base: IR[vect] at +vect (Ch 13) */
00034     k_icu_ier_base = 0x00087200U, /**< ICU.IER[] base: IER[vect/8] bit (vect%8) (Ch 13) */
00035     k_icu_ipr_base = 0x00087300U, /**< ICU.IPR[] base: IPR[vect] priority (Ch 13) */
00036
00037     k_mstpcra_addr = 0x00080010U, /**< MSTPCRA: module stop control A (Ch 11) */
00038     k_prcr_addr    = 0x000803FEU, /**< PRCR: protect register for clock/module stop writes */
00039 } cmt_addr_t;
```

enum icu_cmt0_cfg_t : uint8_t



```
00057             : uint8_t {
00058     k_vect_cmt0       = 28U, /**< ICU vector number for CMT0 CMI0 (Ch 13) */
00059     k_cmt0_priority   = 5U,  /**< IPR priority (0..15); 5 is low-enough for ThreadX tick */
00060     k_icu_ier_cmt0    = 3U,  /**< IER register index for vector 28 (28/8 = 3) */
00061     k_icu_ier_cmt0_bit = 4U, /**< Bit position within IER[3] (28 % 8 = 4) */
00062     k_mstpa_cmt0_bit  = 15U, /**< MSTPCRA bit 15: combined CMT0/CMT1 module stop */
00063 } icu_cmt0_cfg_t;
```

void _tx_timer_interrupt (
                    void) [extern]



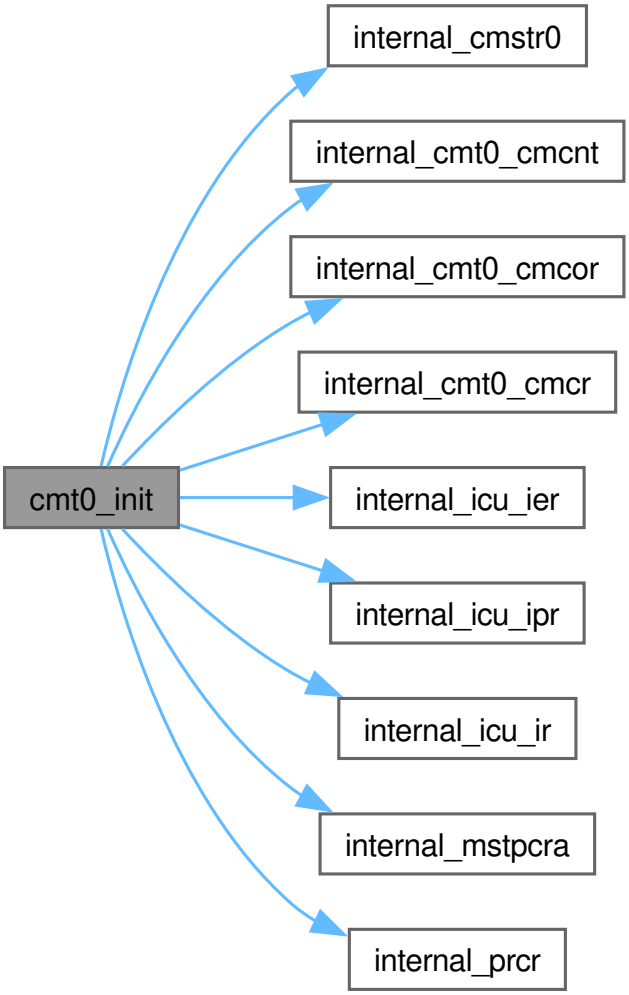
void cmt0_init (
void)

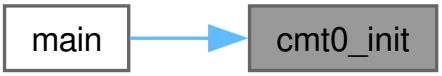
SETPSW I

clock_init()
main()tx_kernel_enter()

cmt0_isr

```
00166 {
00167     /* Release CMT0/CMT1 from module stop (MSTPCRA bit 15 = 0) under PRCR protection. */
00168     *internal_prcr()    = k_prcr_unlock_prc0_prc1;
00169     *internal_mstpcra() &= ~(uint32_t)(1UL « k_mstpa_cmt0_bit);
00170     *internal_prcr()    = k_prcr_lock;
00171
00172     /* Stop CMT0 before reprogramming. */
00173     *internal_cmstr0() &= (uint16_t) ~(uint16_t)k_cmstr0_ch0_bit;
00174
00175     *internal_cmt0_cmcr()  = k_cmt0_cmcr_val;
00176     *internal_cmt0_cmcor() = k_cmt0_cmcor_val;
00177     *internal_cmt0_cmcnt() = 0U;
00178
00179     *internal_icu_ir(k_vect_cmt0)          = 0U;
00180     *internal_icu_ipr(k_vect_cmt0)         = k_cmt0_priority;
00181     *internal_icu_ier(k_icu_ier_cmt0)      |= (uint8_t)(1U « k_icu_ier_cmt0_bit);
00182
00183     *internal_cmstr0() |= k_cmstr0_ch0_bit;
00184
00185     /* Inline asm: RX has no standalone GCC builtin for SETPSW; this is the
00186      * idiomatic way to set PSW.I (global interrupt enable) at runtime. */
00187     ___asm___ volatile("setpsw i");
00188 }
```



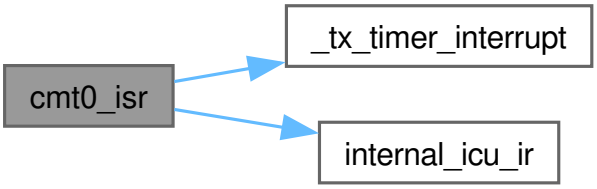


void cmt0_isr (void)

vectors.S___attribute___((interrupt))

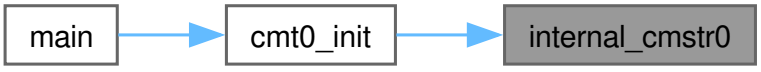
_tx_timer_interrupt()

```
00145 {
00146     *internal_icu_ir(k_vect_cmt0) = 0U;
00147     _tx_timer_interrupt();
00148 }
```



volatile uint16_t * internal_cmstr0 (
void) [inline], [static]

```
00067 {
00068     return (volatile uint16_t *)k_cmstr0_addr;
00069 }
```



volatile uint16_t * internal_cmt0_cmcnt (
void) [inline], [static]

```
00079 {
00080     return (volatile uint16_t *)k_cmt0_cmcnt;
00081 }
```

volatile uint16_t * internal_cmt0_cmcor (
void) [inline], [static]

```
00085 {
00086     return (volatile uint16_t *)k_cmt0_cmcor;
00087 }
```

volatile uint16_t * internal_cmt0_cmcr (
void) [inline], [static]

```
00073 {
00074     return (volatile uint16_t *)k_cmt0_cmcr;
00075 }
```

volatile uint8_t * internal_icu_ier (
uint8_t idx) [inline], [static]

in		

```
00105 {
00106     return (volatile uint8_t *)(k_icu_ier_base + idx);
00107 }
```

volatile uint8_t * internal_icu_ipr (
uint8_t vect) [inline], [static]

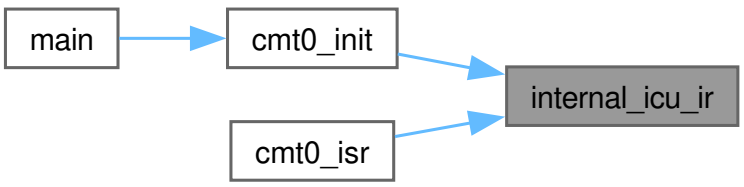
in		

```
00115 {
00116     return (volatile uint8_t *)(k_icu_ipr_base + vect);
00117 }
```

volatile uint8_t * internal_icu_ir (
uint8_t vect) [inline], [static]

in		

```
00095 {
00096     return (volatile uint8_t *)(k_icu_ir_base + vect);
00097 }
```

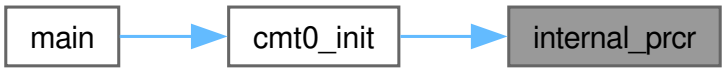


volatile uint32_t * internal_mstpcra (
void) [inline], [static]

```
00121 {
00122     return (volatile uint32_t *)k_mstpcra_addr;
00123 }
```

volatile uint16_t * internal_prcr (
void) [inline], [static]

```
00127 {
00128     return (volatile uint16_t *)k_prcr_addr;
00129 }
```




```
00001 /**
00002  * @file cmt0.c
00003  * @brief CMT0 100 Hz tick source for ThreadX with PCKB = 48 MHz.
00004  *
00005  * @details
00006  * After clock_init() PCKB = 48 MHz. PCKB/32 = 1.5 MHz, divided by 15000
00007  * gives 100 Hz exactly. CMCOR = 14999, CMCR.CKS = 01 (PCKB/32), CMIE = 1.
00008  *
00009  * Vector 28 (CMT0 CMI0) is wired in vectors.S to cmt0_isr below, which is
00010  * marked __attribute__((interrupt)) so GCC generates the RTE epilogue.
00011  *
00012  * Register and bit references follow the RX72N Hardware Manual
00013  * (R01UH0824EJ0111), chapters 13 (ICU), 15 (CMT), and 11 (module stop).
00014  *
00015  * SPDX-License-Identifier: MIT
00016  * @copyright Copyright (c) 2026 Locked Inc.
00017  */
00018
00019 #include <stdint.h>
00020
00021 extern void _tx_timer_interrupt(void);
00022
00023 /**
00024  * @enum cmt_addr_t
00025  * @brief Absolute hardware register addresses used by the CMT0 tick setup.
```

```
00026  */
00027 typedef enum : uintptr_t {
00028     k_cmstr0_addr  = 0x00088000U, /**< CMSTR0: shared CMT0/CMT1 start register (Ch 15) */
00029     k_cmt0_cmcr    = 0x00088002U, /**< CMT0 CMCR: control register (Ch 15) */
00030     k_cmt0_cmcnt   = 0x00088004U, /**< CMT0 CMCNT: counter value (Ch 15) */
00031     k_cmt0_cmcor   = 0x00088006U, /**< CMT0 CMCOR: compare-match target (Ch 15) */
00032
00033     k_icu_ir_base  = 0x00087000U, /**< ICU.IR[] base: IR[vect] at +vect (Ch 13) */
00034     k_icu_ier_base = 0x00087200U, /**< ICU.IER[] base: IER[vect/8] bit (vect%8) (Ch 13) */
00035     k_icu_ipr_base = 0x00087300U, /**< ICU.IPR[] base: IPR[vect] priority (Ch 13) */
00036
00037     k_mstpcra_addr = 0x00080010U, /**< MSTPCRA: module stop control A (Ch 11) */
00038     k_prcr_addr    = 0x000803FEU, /**< PRCR: protect register for clock/module stop writes */
00039 } cmt_addr_t;
00040
00041 /**
00042  * @enum cmt0_cfg_t
00043  * @brief 16-bit configuration values written to CMT0 and PRCR.
00044  */
00045 typedef enum : uint16_t {
00046     k_cmt0_cmcor_val       = 14999U,  /**< (48 MHz / 32) / 100 Hz - 1 = 14999 for 100 Hz tick */
00047     k_cmt0_cmcr_val        = 0x0041U, /**< CMCR: CMIE=1 (interrupt), CKS=01 (PCKB/32) */
00048     k_cmstr0_ch0_bit       = 0x0001U, /**< CMSTR0 bit 0: CMT0 start/stop */
00049     k_prcr_unlock_prc0_prc1 = 0xA503U, /**< Write PRCR=0xA503 to enable PRC0+PRC1 writes */
00050     k_prcr_lock            = 0xA500U, /**< Write PRCR=0xA500 to relock PRC0+PRC1 */
00051 } cmt0_cfg_t;
00052
00053 /**
00054  * @enum icu_cmt0_cfg_t
00055  * @brief ICU vector/priority/bit constants for CMT0 CMI0.
00056  */
00057 typedef enum : uint8_t {
00058     k_vect_cmt0       = 28U, /**< ICU vector number for CMT0 CMI0 (Ch 13) */
00059     k_cmt0_priority   = 5U,  /**< IPR priority (0..15); 5 is low-enough for ThreadX tick */
00060     k_icu_ier_cmt0    = 3U,  /**< IER register index for vector 28 (28/8 = 3) */
00061     k_icu_ier_cmt0_bit = 4U, /**< Bit position within IER[3] (28 % 8 = 4) */
00062     k_mstpa_cmt0_bit  = 15U, /**< MSTPCRA bit 15: combined CMT0/CMT1 module stop */
00063 } icu_cmt0_cfg_t;
00064
00065 /** @brief Typed accessor for CMSTR0 (CMT0/1 start register). */
00066 static inline volatile uint16_t *internal_cmstr0(void)
00067 {
00068     return (volatile uint16_t *)k_cmstr0_addr;
00069 }
00070
00071 /** @brief Typed accessor for CMT0.CMCR. */
00072 static inline volatile uint16_t *internal_cmt0_cmcr(void)
00073 {
00074     return (volatile uint16_t *)k_cmt0_cmcr;
00075 }
00076
00077 /** @brief Typed accessor for CMT0.CMCNT. */
00078 static inline volatile uint16_t *internal_cmt0_cmcnt(void)
00079 {
00080     return (volatile uint16_t *)k_cmt0_cmcnt;
00081 }
00082
00083 /** @brief Typed accessor for CMT0.CMCOR. */
00084 static inline volatile uint16_t *internal_cmt0_cmcor(void)
00085 {
00086     return (volatile uint16_t *)k_cmt0_cmcor;
00087 }
00088
00089 /**
00090  * @brief Typed accessor for ICU.IR[vect] (one-byte interrupt request flag).
00091  * @param[in] vect ICU vector number (0..255).
00092  * @return Volatile pointer to IR[vect].
00093  */
00094 static inline volatile uint8_t *internal_icu_ir(uint8_t vect)
00095 {
00096     return (volatile uint8_t *)(k_icu_ir_base + vect);
00097 }
00098
00099 /**
00100  * @brief Typed accessor for ICU.IER[idx] (vect/8 selects the byte).
00101  * @param[in] idx IER register index (vect / 8).
00102  * @return Volatile pointer to IER[idx].
00103  */
00104 static inline volatile uint8_t *internal_icu_ier(uint8_t idx)
00105 {
00106     return (volatile uint8_t *)(k_icu_ier_base + idx);
00107 }
00108
00109 /**
00110  * @brief Typed accessor for ICU.IPR[vect] (one-byte priority).
00111  * @param[in] vect ICU vector number (0..255).
00112  * @return Volatile pointer to IPR[vect].
```

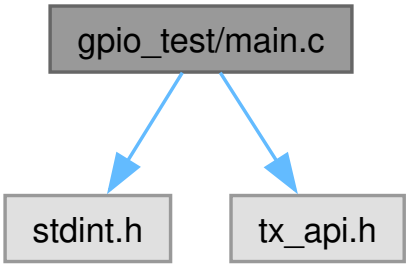
```
00113  */
00114 static inline volatile uint8_t *internal_icu_ipr(uint8_t vect)
00115 {
00116     return (volatile uint8_t *)(k_icu_ipr_base + vect);
00117 }
00118
00119 /** @brief Typed accessor for MSTPCRA (module stop control A). */
00120 static inline volatile uint32_t *internal_mstpcra(void)
00121 {
00122     return (volatile uint32_t *)k_mstpcra_addr;
00123 }
00124
00125 /** @brief Typed accessor for PRCR (clock/module-stop write protect). */
00126 static inline volatile uint16_t *internal_prcr(void)
00127 {
00128     return (volatile uint16_t *)k_prcr_addr;
00129 }
00130
00131 /**
00132  * @brief CMT0 CMI0 ISR: clears IR and drives the ThreadX tick.
00133  *
00134  * @details
00135  * Wired by `vectors.S‘ into slot 28 of the relocatable vector table.
00136  * Marked `___attribute___((interrupt))‘ so GCC-RX emits the RTE prologue
00137  * and epilogue around the body.
00138  *
00139  * @pre Interrupts are globally enabled and this vector is wired in INTB.
00140  * @post `_tx_timer_interrupt()‘ has advanced the ThreadX system clock by
00141  *       exactly one tick.
00142  * @note Executes in ISR context on the ISP.
00143  */
00144 void ___attribute___((interrupt)) cmt0_isr(void)
00145 {
00146     *internal_icu_ir(k_vect_cmt0) = 0U;
00147     _tx_timer_interrupt();
00148 }
00149
00150 /**
00151  * @brief Program CMT0 for a 100 Hz tick and enable its interrupt.
00152  *
00153  * @details
00154  * 1. Releases CMT0 (and CMT1) from module-stop via MSTPCRA bit 15,
00155  *    bracketed by the PRCR unlock/relock protocol.
00156  * 2. Stops CMT0, programs CMCR/CMCOR, clears CMCNT.
00157  * 3. Clears any pending IR, sets IPR=5, enables the vector in IER.
00158  * 4. Starts CMT0 and globally enables interrupts with `SETPSW I‘.
00159  *
00160  * @pre `clock_init()‘ has completed so PCKB = 48 MHz.
00161  * @pre Called from `main()‘ before `tx_kernel_enter()‘.
00162  * @post CMT0 fires at exactly 100 Hz into `cmt0_isr‘.
00163  * @post Global interrupt flag (PSW.I) is set.
00164  */
00165 void cmt0_init(void)
00166 {
00167     /* Release CMT0/CMT1 from module stop (MSTPCRA bit 15 = 0) under PRCR protection. */
00168     *internal_prcr()    = k_prcr_unlock_prc0_prc1;
00169     *internal_mstpcra() &= ~(uint32_t)(1UL « k_mstpa_cmt0_bit);
00170     *internal_prcr()    = k_prcr_lock;
00171
00172     /* Stop CMT0 before reprogramming. */
00173     *internal_cmstr0() &= (uint16_t) ~(uint16_t)k_cmstr0_ch0_bit;
00174
00175     *internal_cmt0_cmcr()  = k_cmt0_cmcr_val;
00176     *internal_cmt0_cmcor() = k_cmt0_cmcor_val;
00177     *internal_cmt0_cmcnt() = 0U;
00178
00179     *internal_icu_ir(k_vect_cmt0)          = 0U;
00180     *internal_icu_ipr(k_vect_cmt0)         = k_cmt0_priority;
00181     *internal_icu_ier(k_icu_ier_cmt0)      |= (uint8_t)(1U « k_icu_ier_cmt0_bit);
00182
00183     *internal_cmstr0() |= k_cmstr0_ch0_bit;
00184
00185     /* Inline asm: RX has no standalone GCC builtin for SETPSW; this is the
00186      * idiomatic way to set PSW.I (global interrupt enable) at runtime. */
00187     ___asm___ volatile(”setpsw i”);
00188 }
```

```
00001 /*
00002  * gpio_test/linker.ld -- Linker script for GPIO breakout board test.
00003  *
00004  * Same memory map as usb_test: 512 KB internal SRAM, 2 MB internal ROM.
00005  * Includes Renesas-syntax sections (P, B, D, C) needed by ThreadX RXv3 port.
```

```
00006 */
00007
00008 MEMORY
00009 {
00010     RAM : ORIGIN = 0x00000004, LENGTH = 0x7FFFC   /* 512 KB - 4 bytes */
00011     ROM : ORIGIN = 0xFFE00000, LENGTH = 0x200000  /* 2 MB             */
00012 }
00013
00014 SECTIONS
00015 {
00016     /* Code -- explicit ROM base so section VMA is anchored correctly. */
00017     .text 0xFFE00000 : AT(0xFFE00000)
00018     {
00019         *(.text*)
00020         *(P)
00021         *(P )
00022         *(.rodata*)
00023         *(C)
00024         *(C )
00025         *(C )
00026         etext = .;
00027     } > ROM
00028
00029     /*
00030      * Relocatable vector table -- startup.S loads INTB with this address.
00031      * Must be 4-byte aligned; only vectors 27 (SWINT) and 28 (CMT0) are
00032      * used, but allocate 256 entries (1024 bytes) for safety.
00033      */
00034     .rvectors ALIGN(4) :
00035     {
00036         __rvectors_start = .;
00037         KEEP(*(.rvectors))
00038         __rvectors_end = .;
00039     } > ROM
00040
00041     /*
00042      * Initialised data -- LMA stays in ROM, VMA in RAM.
00043      * startup.S copies _data..edata from _mdata on boot.
00044      */
00045     _mdata = .;
00046     .data : AT(_mdata)
00047     {
00048         _data = .;
00049         *(.data*)
00050         *(D)
00051         *(D )
00052         *(D )
00053         _edata = .;
00054     } > RAM
00055
00056     /* BSS -- zeroed by startup.S */
00057     .bss :
00058     {
00059         _bss = .;
00060         *(.bss*)
00061         *(COMMON)
00062         *(B)
00063         *(B )
00064         *(B )
00065         _ebss = .;
00066         . = ALIGN(4);
00067         _end = .;
00068     } > RAM AT>RAM
00069
00070     /*
00071      * Stacks -- USP grows down from top of RAM, ISP 4 KB below that.
00072      */
00073     _ustack = ORIGIN(RAM) + LENGTH(RAM);
00074     _istack = _ustack - 0x1000;
00075
00076     /*
00077      * Exception vector table at fixed hardware address 0xFFFFFF80.
00078      * 32 entries -- all unused.
00079      */
00080     .exvectors 0xFFFFFF80 : AT(0xFFFFFF80)
00081     {
00082         LONG(0xFFFFFFFF); LONG(0xFFFFFFFF); LONG(0xFFFFFFFF); LONG(0xFFFFFFFF);
00083         LONG(0xFFFFFFFF); LONG(0xFFFFFFFF); LONG(0xFFFFFFFF); LONG(0xFFFFFFFF);
00084         LONG(0xFFFFFFFF); LONG(0xFFFFFFFF); LONG(0xFFFFFFFF); LONG(0xFFFFFFFF);
00085         LONG(0xFFFFFFFF); LONG(0xFFFFFFFF); LONG(0xFFFFFFFF); LONG(0xFFFFFFFF);
00086         LONG(0xFFFFFFFF); LONG(0xFFFFFFFF); LONG(0xFFFFFFFF); LONG(0xFFFFFFFF);
00087         LONG(0xFFFFFFFF); LONG(0xFFFFFFFF); LONG(0xFFFFFFFF); LONG(0xFFFFFFFF);
00088         LONG(0xFFFFFFFF); LONG(0xFFFFFFFF); LONG(0xFFFFFFFF); LONG(0xFFFFFFFF);
00089         LONG(0xFFFFFFFF); LONG(0xFFFFFFFF); LONG(0xFFFFFFFF);
00090     } > ROM
00091
00092     /* Fixed reset vector at 0xFFFFFFFC */
```

```
00093     .fvectors 0xFFFFFFFC : AT(0xFFFFFFFC)
00094     {
00095         KEEP(*(.fvectors))
00096     } > ROM
00097 }
```

```
#include <stdint.h>
#include "tx_api.h"
```



*

*

*

~

s_pins

s_pins

*

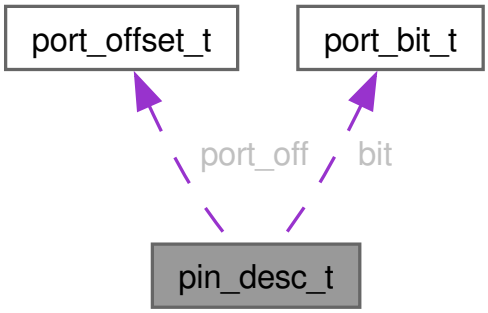
~

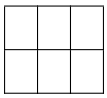
~

~

>>

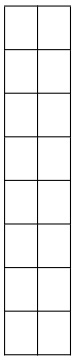
s_pins[]





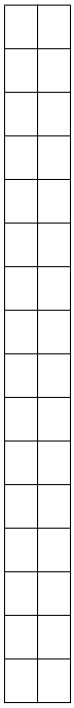
enum port_bit_t : uint8_t

s_pins[]



```
00099             : uint8_t {
00100     k_bit  = 0U, /**< Bit 0 (pin 0) of the port */
00101     k_bit  = 1U, /**< Bit 1 (pin 1) of the port */
00102     k_bit  = 2U, /**< Bit 2 (pin 2) of the port */
00103     k_bit  = 3U, /**< Bit 3 (pin 3) of the port */
00104     k_bit  = 4U, /**< Bit 4 (pin 4) of the port */
00105     k_bit  = 5U, /**< Bit 5 (pin 5) of the port */
00106     k_bit  = 6U, /**< Bit 6 (pin 6) of the port */
00107     k_bit  = 7U, /**< Bit 7 (pin 7) of the port */
00108 } port_bit_t;
```

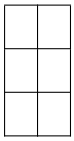
enum port_offset_t : uint8_t



```
00068             : uint8_t {
00069     k_port_off  = 0x00U, /**< Port 0 offset within each register bank */
00070     k_port_off  = 0x01U, /**< Port 1 offset within each register bank */
00071     k_port_off  = 0x02U, /**< Port 2 offset within each register bank */
00072     k_port_off  = 0x03U, /**< Port 3 offset within each register bank */
00073     k_port_off  = 0x04U, /**< Port 4 offset within each register bank */
00074     k_port_off  = 0x05U, /**< Port 5 offset within each register bank */
00075     k_port_off  = 0x06U, /**< Port 6 offset within each register bank */
00076     k_port_off  = 0x07U, /**< Port 7 offset within each register bank */
00077     k_port_off  = 0x08U, /**< Port 8 offset within each register bank */
00078     k_port_off  = 0x09U, /**< Port 9 offset within each register bank */
00079     k_port_off_a = 0x0AU, /**< Port A offset within each register bank */
00080     k_port_off_b = 0x0BU, /**< Port B offset within each register bank */
00081     k_port_off_c = 0x0CU, /**< Port C offset within each register bank */
00082     k_port_off_d = 0x0DU, /**< Port D offset within each register bank */
00083     k_port_off_e = 0x0EU, /**< Port E offset within each register bank */
00084     k_port_off_f = 0x0FU, /**< Port F offset within each register bank */
00085 } port_offset_t;
```

enum port_reg_base_t : uintptr_t

port_offset_t



```
00050             : uintptr_t {
00051     k_pdr_base  = 0x0008C000U, /**< Port direction register bank base */
00052     k_podr_base = 0x0008C020U, /**< Port output data register bank base */
00053     k_pmr_base  = 0x0008C060U, /**< Port mode register bank base */
00054 } port_reg_base_t;
```

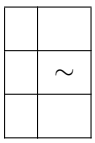
enum rtos_prio_t : uint8_t

```
00198             : uint8_t {
00199     k_sweep_priority = 10U, /**< Priority of the gpio_sweep thread */
00200 } rtos_prio_t;
```

enum rtos_stack_t : uint16_t

```
00206             : uint16_t {
00207     k_sweep_stack_sz = 512U, /**< gpio_sweep thread stack size */
00208 } rtos_stack_t;
```

enum timing_t : uint32_t

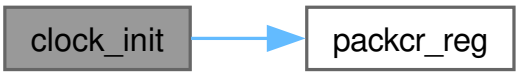


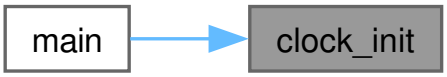
```
00185             : uint32_t {
00186     k_num_pins   = (uint32_t)(sizeof(s_pins) / sizeof(s_pins[0])), /**< Pin count in s_pins */
00187     k_delay_iter = 50000U, /**< nop iterations per HIGH/LOW state (~5 ms at ICLK 96 MHz) */
00188     k_gap_iters  = 500U,   /**< delay() calls in the inter-cycle quiet gap */
00189 } timing_t;
```

void clock_init (
void) [extern]

>>

```
00101 {
00102     volatile rx_system_regs_t *sys = system_regs();
00103
00104     /* Step 0: unlock PRC0 (clock) + PRC1 (module stop) */
00105     *prcr_reg() = k_rx_prcr_unlock_prc0_prc1;
00106
00107     /* Step 1: start HOCO if not already running. */
00108     sys->hococr = k_hococr_running;
00109
00110     /* Step 2: wait for HOCO to stabilise. */
00111     while ((sys->oscovfsr & k_oscovfsr_hcovf) == 0U) {
00112         ___asm___ volatile("nop");
00113     }
00114
00115     /* Step 3: program PLL: HOCO 16 MHz / 1 * 12 = 192 MHz, and enable. */
00116     sys->pllcr  = k_pllcr_hoco_x12;
00117     sys->pllcr2 = k_pllcr2_enable;
00118
00119     /* Step 4: wait for PLL to stabilise. */
00120     while ((sys->oscovfsr & k_oscovfsr_plovf) == 0U) {
00121         ___asm___ volatile("nop");
00122     }
00123
00124     /* Step 5: MANDATORY -- raise flash wait state before ICLK exceeds 120 MHz. */
00125     *memwait_reg() = k_memwait_one;
00126
00127     /* Step 6: program all bus dividers. */
00128     sys->sckcr = k_sckcr_value;
00129
00130     /* Step 7: select main PLL as the UCLK divider source (UPLLSEL=0). */
00131     *packcr_reg() = k_packcr_pll_source;
00132
00133     /* Step 8: program the USB clock divider. */
00134     sys->sckcr2 = k_sckcr2_uck_div4;
00135
00136     /* Step 9: switch the system clock source to PLL. */
00137     sys->sckcr3 = k_sckcr3_cksel_pll;
00138
00139     /* Step 10: relock PRCR. */
00140     *prcr_reg() = k_rx_prcr_lock;
00141 }
```





void cmt0_init (
void) [extern]

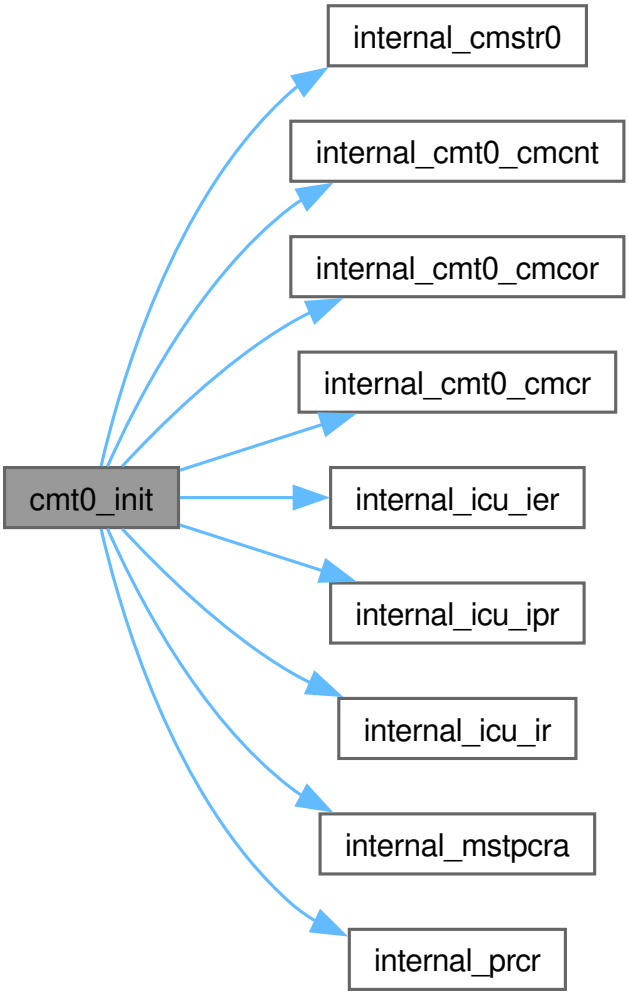
SETPSW I

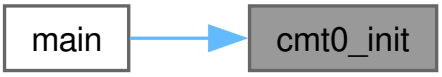
clock_init()

main()tx_kernel_enter()

cmt0_isr

```
00166 {
00167     /* Release CMT0/CMT1 from module stop (MSTPCRA bit 15 = 0) under PRCR protection. */
00168     *internal_prcr()    = k_prcr_unlock_prc0_prc1;
00169     *internal_mstpcra() &= ~(uint32_t)(1UL « k_mstpa_cmt0_bit);
00170     *internal_prcr()    = k_prcr_lock;
00171
00172     /* Stop CMT0 before reprogramming. */
00173     *internal_cmstr0() &= (uint16_t) ~(uint16_t)k_cmstr0_ch0_bit;
00174
00175     *internal_cmt0_cmcr()  = k_cmt0_cmcr_val;
00176     *internal_cmt0_cmcor() = k_cmt0_cmcor_val;
00177     *internal_cmt0_cmcnt() = 0U;
00178
00179     *internal_icu_ir(k_vect_cmt0)          = 0U;
00180     *internal_icu_ipr(k_vect_cmt0)         = k_cmt0_priority;
00181     *internal_icu_ier(k_icu_ier_cmt0)      |= (uint8_t)(1U « k_icu_ier_cmt0_bit);
00182
00183     *internal_cmstr0() |= k_cmstr0_ch0_bit;
00184
00185     /* Inline asm: RX has no standalone GCC builtin for SETPSW; this is the
00186      * idiomatic way to set PSW.I (global interrupt enable) at runtime. */
00187     ___asm___ volatile("setpsw i");
00188 }
```





void internal_delay (
void) [static]

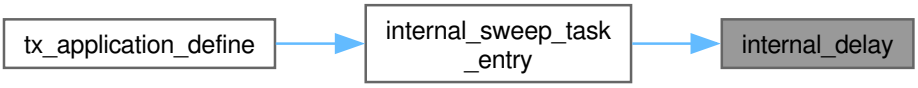
~

tx_thread_sleep~tx_thread_sleep

volatile

k__delay__iter

```
00264 {
00265     for (volatile uint32_t i = 0U; i < k__delay__iter; i++) {
00266         /* Inline asm: emit a real NOP so the RX back-end cannot fold the
00267          * loop body away at higher optimisation levels. `volatile i‘
00268          * alone is not enough on GCC-RX 14.2 when the body is empty.
00269          * Project coding rule requires a justification at each ___asm___ site. */
00270         ___asm___ volatile(”nop”);
00271     }
00272 }
```



void internal__gpio__init__all (
void) [static]

s__pins

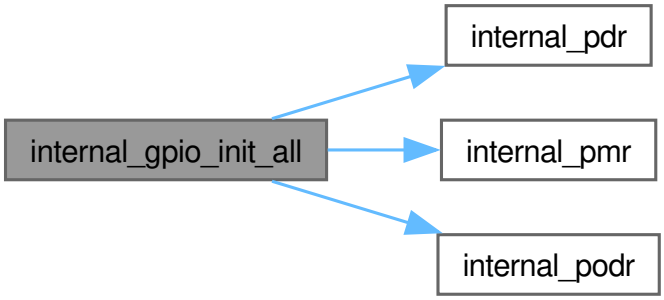
clock__init()

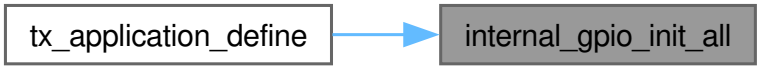
s__pinsconst

s__pins

tx__application__define

```
00289 {
00290     for (uint32_t i = 0U; i < k__num__pins; i++) {
00291         uint8_t off  = (uint8_t)s__pins[i].port__off;
00292         uint8_t mask = (uint8_t)(1U « (uint8_t)s__pins[i].bit);
00293         *internal__pmr(off)  &= (uint8_t)~mask;
00294         *internal__pdr(off)  |= mask;
00295         *internal__podr(off) &= (uint8_t)~mask;
00296     }
00297 }
```

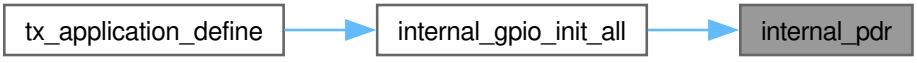




volatile uint8_t * internal_pdr (
uint8_t port_off) [inline], [static]

in		port_offset_t

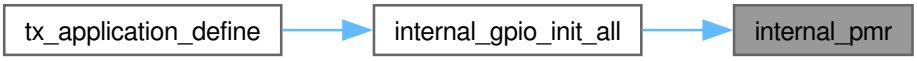
```
00226 {
00227     return (volatile uint8_t *)(k_pdr_base + port_off);
00228 }
```



volatile uint8_t * internal_pmr (
uint8_t port_off) [inline], [static]

in		port_offset_t

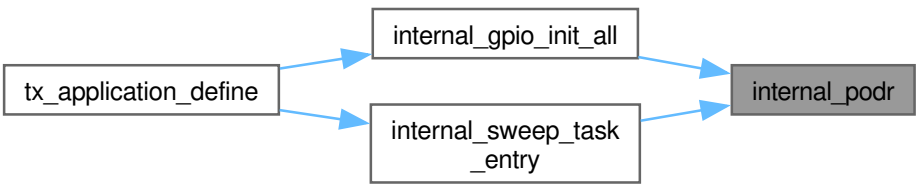
```
00246 {
00247     return (volatile uint8_t *)(k_pmr_base + port_off);
00248 }
```



volatile uint8_t * internal_podr (
uint8_t port_off) [inline], [static]

in		port_offset_t

```
00236 {
00237     return (volatile uint8_t *)(k_podr_base + port_off);
00238 }
```



void internal_sweep_task_entry (
ULONG arg) [static]

s_pins

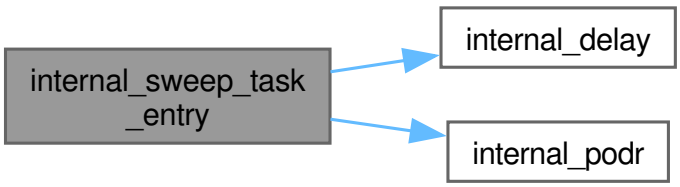
k_gap_itersdelay()

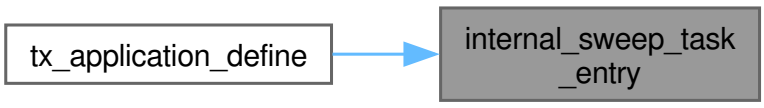
in		

internal_gpio_init_all()

s_pins

```
00315 {
00316     (void)arg;
00317
00318     for (;;) {
00319         for (uint32_t i = 0U; i < k_num_pins; i++) {
00320             uint8_t off  = (uint8_t)s_pins[i].port_off;
00321             uint8_t mask = (uint8_t)(1U << (uint8_t)s_pins[i].bit);
00322             *internal_podr(off) |= mask;
00323             internal_delay();
00324             *internal_podr(off) &= (uint8_t)~mask;
00325             internal_delay();
00326         }
00327         for (uint32_t g = 0U; g < k_gap_iters; g++) {
00328             internal_delay();
00329         }
00330     }
00331 }
```





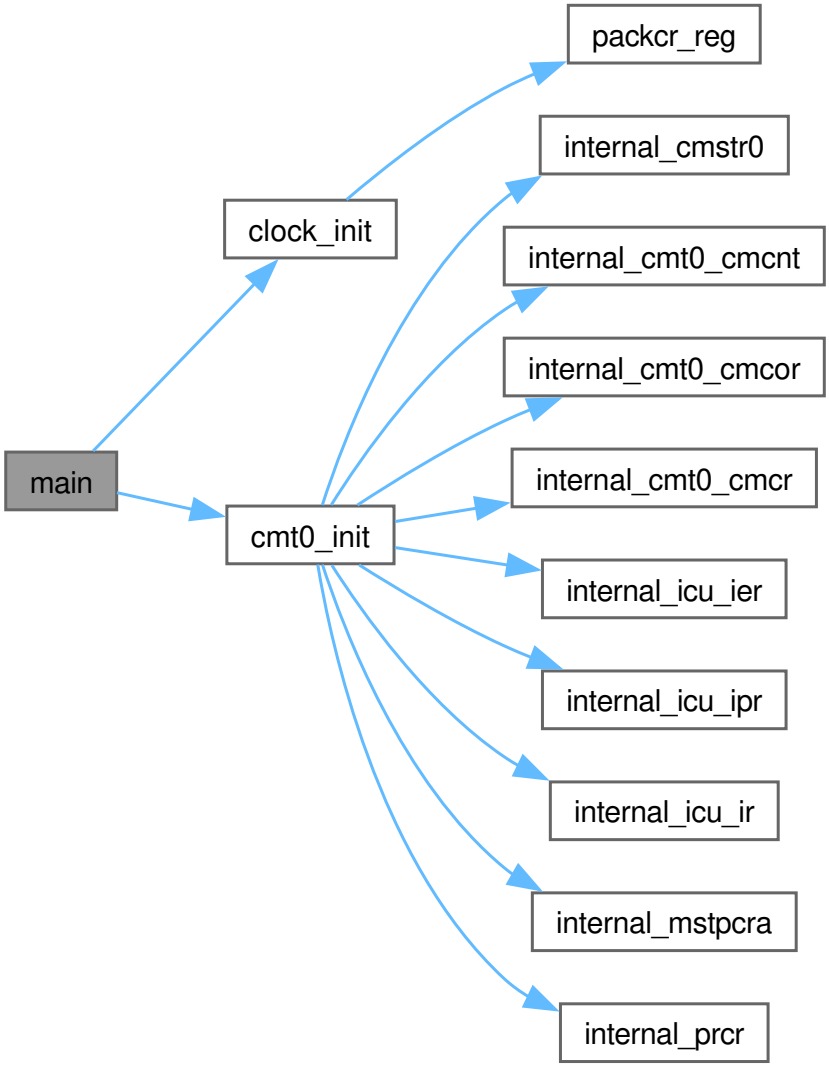
int main (
                    void)

clock_init()cmt0_init()cmt0_init()tx_kernel_enter()

return

startup.S
_PowerON_Reset_PC

```
00401 {
00402     clock_init();
00403     cmt0_init();
00404     tx_kernel_enter();
00405     return 0;
00406 }
```



void tx_application_define (
void * first_unused_memory)

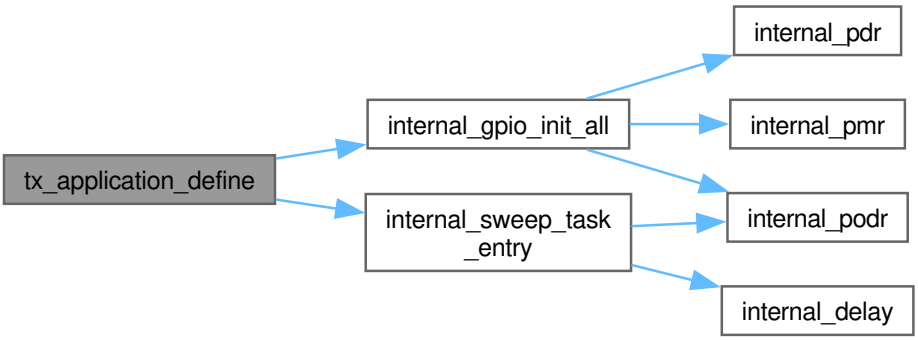
tx_kernel_enter()tx_thread_create

in		

tx_kernel_enter()

s_pins

```
00353 {
00354     (void)first_unused_memory;
00355
00356     internal_gpio_init_all();
00357
00358     UINT status = tx_thread_create(&s_sweep_tcb, "gpio_sweep",
00359                                    internal_sweep_task_entry, 0U,
00360                                    s_sweep_stack, k_sweep_stack_sz,
00361                                    k_sweep_priority, k_sweep_priority,
00362                                    TX_NO_TIME_SLICE, TX_AUTO_START);
00363     if (status != TX_SUCCESS) {
00364         /* Park here so the bench observes a flat line instead of a
00365          * ghost sweep. Any failure (TX_SIZE_ERROR / TX_PTR_ERROR /
00366          * TX_START_ERROR) surfaces loudly as "no edges ever" which
00367          * is easy to spot on the logic analyser. */
00368         for (;;) {
00369             __asm__ volatile("nop");
00370         }
00371     }
00372 }
```



const pin_desc_t s_pins[] [static]

```
00135                         {
00136     { k_port_off, k_bit  }, { k_port_off, k_bit  }, { k_port_off, k_bit  },
00137     { k_port_off, k_bit  }, { k_port_off, k_bit  }, { k_port_off, k_bit  },
00138     { k_port_off, k_bit  }, { k_port_off, k_bit  }, { k_port_off, k_bit  },
00139     { k_port_off, k_bit  }, { k_port_off, k_bit  },
00140     { k_port_off, k_bit  }, { k_port_off, k_bit  }, { k_port_off, k_bit  },
00141     { k_port_off, k_bit  }, { k_port_off, k_bit  }, { k_port_off, k_bit  },
00142     { k_port_off, k_bit  }, { k_port_off, k_bit  },
00143     { k_port_off, k_bit  }, { k_port_off, k_bit  }, { k_port_off, k_bit  },
00144     { k_port_off, k_bit  }, { k_port_off, k_bit  }, { k_port_off, k_bit  },
00145     { k_port_off, k_bit  }, { k_port_off, k_bit  },
00146     { k_port_off, k_bit  }, { k_port_off, k_bit  }, { k_port_off, k_bit  },
00147     { k_port_off, k_bit  }, { k_port_off, k_bit  }, { k_port_off, k_bit  },
00148     { k_port_off, k_bit  },
00149     { k_port_off, k_bit  }, { k_port_off, k_bit  }, { k_port_off, k_bit  },
00150     { k_port_off, k_bit  }, { k_port_off, k_bit  }, { k_port_off, k_bit  },
00151     { k_port_off, k_bit  }, { k_port_off, k_bit  },
00152     { k_port_off, k_bit  }, { k_port_off, k_bit  }, { k_port_off, k_bit  },
00153     { k_port_off, k_bit  }, { k_port_off, k_bit  }, { k_port_off, k_bit  },
00154     { k_port_off, k_bit  }, { k_port_off, k_bit  },
00155     { k_port_off, k_bit  }, { k_port_off, k_bit  }, { k_port_off, k_bit  },
00156     { k_port_off, k_bit  }, { k_port_off, k_bit  }, { k_port_off, k_bit  },
00157     { k_port_off, k_bit  }, { k_port_off, k_bit  }, { k_port_off, k_bit  },
00158     { k_port_off, k_bit  },
00159     { k_port_off_a, k_bit  }, { k_port_off_a, k_bit  }, { k_port_off_a, k_bit  },
00160     { k_port_off_a, k_bit  }, { k_port_off_a, k_bit  }, { k_port_off_a, k_bit  },
00161     { k_port_off_a, k_bit  }, { k_port_off_a, k_bit  },
```

```
00162     { k_port_off_b, k_bit  }, { k_port_off_b, k_bit  }, { k_port_off_b, k_bit  },
00163     { k_port_off_b, k_bit  }, { k_port_off_b, k_bit  }, { k_port_off_b, k_bit  },
00164     { k_port_off_c, k_bit  }, { k_port_off_c, k_bit  }, { k_port_off_c, k_bit  },
00165     { k_port_off_c, k_bit  }, { k_port_off_c, k_bit  }, { k_port_off_c, k_bit  },
00166     { k_port_off_c, k_bit  },
00167     { k_port_off_d, k_bit  }, { k_port_off_d, k_bit  }, { k_port_off_d, k_bit  },
00168     { k_port_off_d, k_bit  }, { k_port_off_d, k_bit  }, { k_port_off_d, k_bit  },
00169     { k_port_off_d, k_bit  }, { k_port_off_d, k_bit  },
00170     { k_port_off_e, k_bit  }, { k_port_off_e, k_bit  }, { k_port_off_e, k_bit  },
00171     { k_port_off_e, k_bit  }, { k_port_off_e, k_bit  }, { k_port_off_e, k_bit  },
00172     { k_port_off_e, k_bit  }, { k_port_off_e, k_bit  },
00173     { k_port_off_f, k_bit  },
00174 };
```

uint8_t s_sweep_stack[k_sweep_stack_sz] [static]

TX_THREAD s_sweep_tcb [static]

```
00001 /**
00002  * @file main.c
00003  * @brief GPIO breakout verification firmware for RX72N, ThreadX edition.
00004  *
00005  * @details
00006  * Mirrors the structure that works in blinky_rtos: one ThreadX thread that
00007  * drives the pins with a busy-wait delay (NOT tx_thread_sleep, which stalls
00008  * when the scheduler/tick isn't ticking for any reason). A single 100 Hz
00009  * CMT0 tick keeps the kernel happy even though we do not use it for
00010  * sleep-based timing.
00011  *
00012  * Test pattern per pin:
00013  *   1. Drive HIGH for ~5 ms (busy-wait loop)
00014  *   2. Drive LOW  for ~5 ms
00015  *   3. Advance to next pin
00016  *
00017  * After all pins are tested a long quiet gap (~2.3 s) separates cycles so
00018  * the host script can anchor capture timing to the gap.
00019  *
00020  * Clock path:
00021  *   clock_init() brings us from HOCO 16 MHz -> PLL 192 MHz -> ICLK 96 MHz
00022  *   / PCKB 48 MHz. CMT0 is then programmed at PCKB/32 for 100 Hz tick.
00023  *
00024  * PJ3 and PJ5 are excluded: on RX72N they double as JTAG TMS/TDO and driving
00025  * them while the E2 Lite has the debug interface latched hangs the MCU.
00026  *
00027  * SPDX-License-Identifier: MIT
00028  * @copyright Copyright (c) 2026 Locked Inc.
00029  */
00030
00031 #include <stdint.h>
00032 #include "tx_api.h"
00033
00034 /*
      ============================================================================
00035  * GPIO register base addresses (RX72N Hardware Manual Ch 22)
00036  *
      ============================================================================
      */
00037 /**
00038  * @enum port_reg_base_t
00039  * @brief Base addresses of the per-port GPIO register banks.
00040  *
00041  * @details
00042  * Each GPIO port has three 8-bit registers indexed by `port_offset_t`:
00043  *   - PDR  (direction):  0 = input,  1 = output
00044  *   - PODR (output data): 0 = LOW,   1 = HIGH
00045  *   - PMR  (mode):       0 = GPIO,  1 = peripheral
00046  * Address = base + port_offset.
00047  *
```

```
00048  * @see RX72N Hardware Manual, Chapter 22 (I/O Ports)
00049  */
00050 typedef enum : uintptr_t {
00051     k_pdr_base  = 0x0008C000U, /**< Port direction register bank base */
00052     k_podr_base = 0x0008C020U, /**< Port output data register bank base */
00053     k_pmr_base  = 0x0008C060U, /**< Port mode register bank base */
00054 } port_reg_base_t;
00055
00056 /*
       ================================================================================
00057  * Port offsets (PDR[k_port_off_x] = PDR_base + offset)
00058  *
       ================================================================================
       */
00059 /**
00060  * @enum port_offset_t
00061  * @brief Per-port index added to every GPIO register bank base.
00062  *
00063  * @details
00064  * Ports 0..9 map to offsets 0x00..0x09, ports A..F to 0x0A..0x0F. Port J
00065  * lives at offset 0x12 because offsets 0x10 (Port G) and 0x11 (Port H) are
00066  * reserved on this package even though the address space is allocated.
00067  */
00068 typedef enum : uint8_t {
00069     k_port_off  = 0x00U, /**< Port 0 offset within each register bank */
00070     k_port_off  = 0x01U, /**< Port 1 offset within each register bank */
00071     k_port_off  = 0x02U, /**< Port 2 offset within each register bank */
00072     k_port_off  = 0x03U, /**< Port 3 offset within each register bank */
00073     k_port_off  = 0x04U, /**< Port 4 offset within each register bank */
00074     k_port_off  = 0x05U, /**< Port 5 offset within each register bank */
00075     k_port_off  = 0x06U, /**< Port 6 offset within each register bank */
00076     k_port_off  = 0x07U, /**< Port 7 offset within each register bank */
00077     k_port_off  = 0x08U, /**< Port 8 offset within each register bank */
00078     k_port_off  = 0x09U, /**< Port 9 offset within each register bank */
00079     k_port_off_a = 0x0AU, /**< Port A offset within each register bank */
00080     k_port_off_b = 0x0BU, /**< Port B offset within each register bank */
00081     k_port_off_c = 0x0CU, /**< Port C offset within each register bank */
00082     k_port_off_d = 0x0DU, /**< Port D offset within each register bank */
00083     k_port_off_e = 0x0EU, /**< Port E offset within each register bank */
00084     k_port_off_f = 0x0FU, /**< Port F offset within each register bank */
00085 } port_offset_t;
00086
00087 /*
       ================================================================================
00088  * Port bit positions (0..7 within each 8-bit PDR/PODR/PMR register)
00089  *
       ================================================================================
       */
00090 /**
00091  * @enum port_bit_t
00092  * @brief Bit position within an 8-bit port register.
00093  *
00094  * @details
00095  * RX72N ports are 8 bits wide; pin N of a given port is bit N of the port's
00096  * PDR/PODR/PMR. Using a typed enum in `s_pins[]` keeps the table readable
00097  * and satisfies the STAR coding rule against raw numeric literals.
00098  */
00099 typedef enum : uint8_t {
00100     k_bit  = 0U, /**< Bit 0 (pin 0) of the port */
00101     k_bit  = 1U, /**< Bit 1 (pin 1) of the port */
00102     k_bit  = 2U, /**< Bit 2 (pin 2) of the port */
00103     k_bit  = 3U, /**< Bit 3 (pin 3) of the port */
00104     k_bit  = 4U, /**< Bit 4 (pin 4) of the port */
00105     k_bit  = 5U, /**< Bit 5 (pin 5) of the port */
00106     k_bit  = 6U, /**< Bit 6 (pin 6) of the port */
00107     k_bit  = 7U, /**< Bit 7 (pin 7) of the port */
00108 } port_bit_t;
00109
00110 /*
       ================================================================================
00111  * Pin descriptor
00112  *
       ================================================================================
       */
00113 /**
00114  * @struct pin_desc_t
00115  * @brief One entry in the firmware's pin sweep table.
00116  *
00117  * @details
00118  * The sweep iterates `s_pins[]` in order; its index is the firmware pin
00119  * index matched by the host capture script.
00120  */
00121 typedef struct {
00122     port_offset_t port_off; /**< Port offset, used with k_pdr_base/k_podr_base/k_pmr_base */
00123     port_bit_t    bit;      /**< Bit position (0..7) within the 8-bit port register */
00124 } pin_desc_t;
00125
```

```
00126 /*
      ================================================================================
00127  * Complete pin table. Array index = firmware sweep index. BGA pin numbers
00128  * from `pin_map.md` are deliberately NOT encoded here -- the firmware only
00129  * cares about port/bit. PJ3 and PJ5 are intentionally omitted: on RX72N
00130  * they double as JTAG TMS/TDO; driving them while the E2 Lite holds the
00131  * debug interface hangs the MCU (empirically observed, every pin after
00132  * the first PJ write went silent).
00133  *
      ================================================================================
      */
00134 /** @brief Full sweep table, one entry per tested GPIO pin. */
00135 static const pin_desc_t s_pins[] = {
00136     { k_port_off, k_bit  }, { k_port_off, k_bit  }, { k_port_off, k_bit  },
00137     { k_port_off, k_bit  }, { k_port_off, k_bit  }, { k_port_off, k_bit  },
00138     { k_port_off, k_bit  }, { k_port_off, k_bit  }, { k_port_off, k_bit  },
00139     { k_port_off, k_bit  }, { k_port_off, k_bit  },
00140     { k_port_off, k_bit  }, { k_port_off, k_bit  }, { k_port_off, k_bit  },
00141     { k_port_off, k_bit  }, { k_port_off, k_bit  }, { k_port_off, k_bit  },
00142     { k_port_off, k_bit  }, { k_port_off, k_bit  },
00143     { k_port_off, k_bit  }, { k_port_off, k_bit  }, { k_port_off, k_bit  },
00144     { k_port_off, k_bit  }, { k_port_off, k_bit  }, { k_port_off, k_bit  },
00145     { k_port_off, k_bit  }, { k_port_off, k_bit  },
00146     { k_port_off, k_bit  }, { k_port_off, k_bit  }, { k_port_off, k_bit  },
00147     { k_port_off, k_bit  }, { k_port_off, k_bit  }, { k_port_off, k_bit  },
00148     { k_port_off, k_bit  },
00149     { k_port_off, k_bit  }, { k_port_off, k_bit  }, { k_port_off, k_bit  },
00150     { k_port_off, k_bit  }, { k_port_off, k_bit  }, { k_port_off, k_bit  },
00151     { k_port_off, k_bit  }, { k_port_off, k_bit  },
00152     { k_port_off, k_bit  }, { k_port_off, k_bit  }, { k_port_off, k_bit  },
00153     { k_port_off, k_bit  }, { k_port_off, k_bit  }, { k_port_off, k_bit  },
00154     { k_port_off, k_bit  }, { k_port_off, k_bit  },
00155     { k_port_off, k_bit  }, { k_port_off, k_bit  }, { k_port_off, k_bit  },
00156     { k_port_off, k_bit  }, { k_port_off, k_bit  }, { k_port_off, k_bit  },
00157     { k_port_off, k_bit  }, { k_port_off, k_bit  }, { k_port_off, k_bit  },
00158     { k_port_off, k_bit  },
00159     { k_port_off_a, k_bit  }, { k_port_off_a, k_bit  }, { k_port_off_a, k_bit  },
00160     { k_port_off_a, k_bit  }, { k_port_off_a, k_bit  }, { k_port_off_a, k_bit  },
00161     { k_port_off_a, k_bit  }, { k_port_off_a, k_bit  },
00162     { k_port_off_b, k_bit  }, { k_port_off_b, k_bit  }, { k_port_off_b, k_bit  },
00163     { k_port_off_b, k_bit  }, { k_port_off_b, k_bit  }, { k_port_off_b, k_bit  },
00164     { k_port_off_c, k_bit  }, { k_port_off_c, k_bit  }, { k_port_off_c, k_bit  },
00165     { k_port_off_c, k_bit  }, { k_port_off_c, k_bit  }, { k_port_off_c, k_bit  },
00166     { k_port_off_c, k_bit  },
00167     { k_port_off_d, k_bit  }, { k_port_off_d, k_bit  }, { k_port_off_d, k_bit  },
00168     { k_port_off_d, k_bit  }, { k_port_off_d, k_bit  }, { k_port_off_d, k_bit  },
00169     { k_port_off_d, k_bit  }, { k_port_off_d, k_bit  },
00170     { k_port_off_e, k_bit  }, { k_port_off_e, k_bit  }, { k_port_off_e, k_bit  },
00171     { k_port_off_e, k_bit  }, { k_port_off_e, k_bit  }, { k_port_off_e, k_bit  },
00172     { k_port_off_e, k_bit  }, { k_port_off_e, k_bit  },
00173     { k_port_off_f, k_bit  },
00174 };
00175
00176 /*
      ================================================================================
00177  * Timing constants. Busy-wait delay at ICLK = 96 MHz. The actual pin
00178  * period drifts a little with optimisation level; gpio_verify.py measures
00179  * it empirically from the capture.
00180  *
      ================================================================================
      */
00181 /**
00182  * @enum timing_t
00183  * @brief Sweep loop iteration and gap counts.
00184  */
00185 typedef enum : uint32_t {
00186     k_num_pins   = (uint32_t)(sizeof(s_pins) / sizeof(s_pins[0])), /**< Pin count in s_pins */
00187     k_delay_iter = 50000U, /**< nop iterations per HIGH/LOW state (~5 ms at ICLK 96 MHz) */
00188     k_gap_iters  = 500U,   /**< delay() calls in the inter-cycle quiet gap */
00189 } timing_t;
00190
00191 /*
      ================================================================================
00192  * ThreadX objects
00193  *
      ================================================================================
      */
00194 /**
00195  * @enum rtos_prio_t
00196  * @brief ThreadX priority numbers used by this firmware.
00197  */
00198 typedef enum : uint8_t {
00199     k_sweep_priority = 10U, /**< Priority of the gpio_sweep thread */
00200 } rtos_prio_t;
00201
00202 /**
00203  * @enum rtos_stack_t
```

```
00204  * @brief ThreadX stack sizes in bytes.
00205  */
00206 typedef enum : uint16_t {
00207     k_sweep_stack_sz = 512U, /**< gpio_sweep thread stack size */
00208 } rtos_stack_t;
00209
00210 /** @var s_sweep_tcb @brief ThreadX control block for the gpio_sweep task. */
00211 static TX_THREAD s_sweep_tcb;
00212
00213 /** @var s_sweep_stack @brief Static stack backing s_sweep_tcb. */
00214 static uint8_t   s_sweep_stack[k_sweep_stack_sz];
00215
00216 /*
      ============================================================================
00217  * Inline accessors
00218  *
      ============================================================================
      */
00219
00220 /**
00221  * @brief Pointer to the PDR (direction) register for a given port offset.
00222  * @param[in] port_off Port offset constant from `port_offset_t`.
00223  * @return Volatile pointer to the 8-bit PDR register for that port.
00224  */
00225 static inline volatile uint8_t *internal_pdr(uint8_t port_off)
00226 {
00227     return (volatile uint8_t *)(k_pdr_base + port_off);
00228 }
00229
00230 /**
00231  * @brief Pointer to the PODR (output data) register for a given port offset.
00232  * @param[in] port_off Port offset constant from `port_offset_t`.
00233  * @return Volatile pointer to the 8-bit PODR register for that port.
00234  */
00235 static inline volatile uint8_t *internal_podr(uint8_t port_off)
00236 {
00237     return (volatile uint8_t *)(k_podr_base + port_off);
00238 }
00239
00240 /**
00241  * @brief Pointer to the PMR (mode) register for a given port offset.
00242  * @param[in] port_off Port offset constant from `port_offset_t`.
00243  * @return Volatile pointer to the 8-bit PMR register for that port.
00244  */
00245 static inline volatile uint8_t *internal_pmr(uint8_t port_off)
00246 {
00247     return (volatile uint8_t *)(k_pmr_base + port_off);
00248 }
00249
00250 /**
00251  * @brief Busy-wait for one pin state (~5 ms at ICLK 96 MHz).
00252  *
00253  * @details
00254  * Used in place of `tx_thread_sleep`: the 100 Hz ThreadX tick is too coarse
00255  * for a ~5 ms pulse, and the earlier `tx_thread_sleep`-based version stalled
00256  * the sweep thread indefinitely (same pattern blinky_rtos avoids for its
00257  * sub-tick timing).
00258  *
00259  * @pre ICLK is running at 96 MHz (clock_init has completed).
00260  * @pre The compiler is honouring `volatile` on the loop counter.
00261  * @post `k_delay_iter` nop instructions have retired and control returns.
00262  */
00263 static void internal_delay(void)
00264 {
00265     for (volatile uint32_t i = 0U; i < k_delay_iter; i++) {
00266         /* Inline asm: emit a real NOP so the RX back-end cannot fold the
00267          * loop body away at higher optimisation levels. `volatile i`
00268          * alone is not enough on GCC-RX 14.2 when the body is empty.
00269          * Project coding rule requires a justification at each ___asm___ site. */
00270         ___asm___ volatile("nop");
00271     }
00272 }
00273
00274 /**
00275  * @brief Configure every pin in `s_pins` as a GPIO output driven LOW.
00276  *
00277  * @details
00278  * For each entry: clears the PMR bit (GPIO mode, not peripheral), sets the
00279  * PDR bit (output direction), and clears the PODR bit (LOW). Default RX72N
00280  * reset state is PMR=0 across the board, so the PMR clear is defensive.
00281  *
00282  * @pre `clock_init()` has run so PCKB is stable when the writes retire.
00283  * @pre `s_pins` is a `const` table in ROM and is safe to read here.
00284  * @post Every pin in `s_pins` has PMR=0, PDR=1, PODR=0.
00285  * @post No other port bits were modified (read-modify-write per bit).
00286  * @note Called exactly once, from `tx_application_define`.
00287  */
```

```
00288 static void internal_gpio_init_all(void)
00289 {
00290     for (uint32_t i = 0U; i < k_num_pins; i++) {
00291         uint8_t off  = (uint8_t)s_pins[i].port_off;
00292         uint8_t mask = (uint8_t)(1U << (uint8_t)s_pins[i].bit);
00293         *internal_pmr(off)  &= (uint8_t)~mask;
00294         *internal_pdr(off)  |= mask;
00295         *internal_podr(off) &= (uint8_t)~mask;
00296     }
00297 }
00298
00299 /**
00300  * @brief Sweep task: pulses each pin in `s_pins` order, forever.
00301  *
00302  * @details
00303  * Each iteration drives one pin HIGH, busy-waits, drives it LOW,
00304  * busy-waits, moves to the next entry. After every pin has pulsed once
00305  * the thread busy-waits through `k_gap_iters` additional `delay()` calls
00306  * so the host capture script can anchor on the quiet gap.
00307  *
00308  * @param[in] arg Unused ThreadX thread argument.
00309  * @pre `internal_gpio_init_all()` has completed.
00310  * @pre The ThreadX kernel is running (scheduler + tick are up).
00311  * @post Never returns; the loop drives the full `s_pins` cycle exactly
00312  *       once per iteration.
00313  */
00314 static void internal_sweep_task_entry(ULONG arg)
00315 {
00316     (void)arg;
00317
00318     for (;;) {
00319         for (uint32_t i = 0U; i < k_num_pins; i++) {
00320             uint8_t off  = (uint8_t)s_pins[i].port_off;
00321             uint8_t mask = (uint8_t)(1U << (uint8_t)s_pins[i].bit);
00322             *internal_podr(off) |= mask;
00323             internal_delay();
00324             *internal_podr(off) &= (uint8_t)~mask;
00325             internal_delay();
00326         }
00327         for (uint32_t g = 0U; g < k_gap_iters; g++) {
00328             internal_delay();
00329         }
00330     }
00331 }
00332
00333 /*
    ============================================================================
00334  * tx_application_define -- called by ThreadX from tx_kernel_enter().
00335  *
    ============================================================================
    */
00336 /**
00337  * @brief ThreadX hook: configure GPIO and start the sweep task.
00338  *
00339  * @details
00340  * ThreadX invokes this from inside `tx_kernel_enter()` once the kernel
00341  * is initialised. If `tx_thread_create` fails the function parks in an
00342  * infinite loop so the bench observes a flat line instead of a silently
00343  * missing sweep (which would otherwise look identical to a crashed
00344  * thread on the logic analyser).
00345  *
00346  * @param[in] first_unused_memory Unused; kernel-provided heap anchor.
00347  * @pre The ThreadX kernel has been initialised by `tx_kernel_enter()`.
00348  * @post All `s_pins` are GPIO outputs driven LOW.
00349  * @post Either the gpio_sweep thread is created (success) or control is
00350  *       parked in an infinite nop loop (failure).
00351  */
00352 void tx_application_define(void *first_unused_memory)
00353 {
00354     (void)first_unused_memory;
00355
00356     internal_gpio_init_all();
00357
00358     UINT status = tx_thread_create(&s_sweep_tcb, "gpio_sweep",
00359                                    internal_sweep_task_entry, 0U,
00360                                    s_sweep_stack, k_sweep_stack_sz,
00361                                    k_sweep_priority, k_sweep_priority,
00362                                    TX_NO_TIME_SLICE, TX_AUTO_START);
00363     if (status != TX_SUCCESS) {
00364         /* Park here so the bench observes a flat line instead of a
00365          * ghost sweep. Any failure (TX_SIZE_ERROR / TX_PTR_ERROR /
00366          * TX_START_ERROR) surfaces loudly as "no edges ever" which
00367          * is easy to spot on the logic analyser. */
00368         for (;;) {
00369             __asm__ volatile("nop");
00370         }
00371     }
```

```
00372 }
00373
00374 /*
      ============================================================================
00375  * External init hooks
00376  *
      ============================================================================
      */
00377 extern void clock_init(void); /**< HOCO 16 MHz -> PLL 192 MHz -> ICLK 96 MHz */
00378 extern void cmt0_init(void);  /**< 100 Hz CMT0 tick source for ThreadX */
00379
00380 /*
      ============================================================================
00381  * main
00382  *
      ============================================================================
      */
00383 /**
00384  * @brief Firmware entry point.
00385  *
00386  * @details
00387  * Brings up the PLL (96 MHz), arms the CMT0 100 Hz tick, then hands off
00388  * to ThreadX. Order matters: `clock_init()` must run first so PCKB is
00389  * 48 MHz before `cmt0_init()` programs the CMT registers; `cmt0_init()`
00390  * runs before `tx_kernel_enter()` as in blinky_rtos.
00391  *
00392  * @return Nominally 0, but control never reaches the `return`: the
00393  *         ThreadX scheduler is the final owner of CPU time.
00394  * @retval 0 Unreachable.
00395  * @pre Called from `startup.S` with the user/interrupt stack pointers
00396  *      valid and BSS zeroed.
00397  * @pre The RX72N reset vector resolves to `__PowerON_Reset_PC`.
00398  * @post Never returns; ThreadX scheduler owns control.
00399  */
00400 int main(void)
00401 {
00402     clock_init();
00403     cmt0_init();
00404     tx_kernel_enter();
00405     return 0;
00406 }
```